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A New Control Strategy for Electric Power Steering on Low Friction Roads

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ABSTRACT

In vehicles equipped with conventional Electric Power Steering (EPS) systems, the steering effort felt by the driver can be unreasonably low when driving on slippery roads. This may lead inexperienced drivers to steer more than what is required in a turn and risk losing control of the vehicle. Thus, it is sensible for tire-road friction to be accounted for in the design of future EPS systems. This paper describes the design of an auxiliary EPS controller that manipulates torque delivery of current EPS systems by supplying its motor with a compensation current controlled by a fuzzy logic algorithm that considers tire-road friction among other factors. Moreover, a steering system model, a nonlinear vehicle dynamics model and a Dugoff tire model are developed in MATLAB/Simulink. Physical testing is conducted to validate the virtual model and confirm that steering torque decreases considerably on low friction roads. Finally, simulation results show that the new EPS assist control strategy successfully increases steering effort under slippery conditions without compromising ease of steering, while ensuring that road feel also increases with decreasing in road friction and/or increasing vehicle speed as an additional safety measure.

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INTRODUCTION

Recent developments in Electric Power Steering (EPS) have triggered much attention from automakers and the automotive research community. The energy efficiency, structural simplicity, low environmental impact and excellent handling performance of the EPS have made it an appealing alternative to Hydraulic Power Steering (HPS) systems [1, 2, 3, 4].

When modeling conventional EPS systems, tire characteristics are often made linear, such that the self-aligning torque transmitted to the steering column from the front wheels is assumed to be directly proportional to the steering angle of the front wheels [4, 5]. However, this approximation is not suitable under all circumstances, e.g. when information such as variation in road friction needs to be conveyed to the driver. It is well-known that alignment torque increases with increasing sideslip angle when tire characteristics are within the linear region, but decreases sharply beyond this region. On high friction coefficient roads, this linear region is sufficiently large such that the tire slip angle rarely exceeds it, but such occurrence is more likely on low friction coefficient roads, where the linear region is much narrower [6]. When the linear region is exceeded, as often occurs under slippery conditions, steering effort decreases considerably, resulting in a loss of road feel. This problem can be addressed using a new EPS control strategy.

Much work has already been done on improving the handling performance of EPS systems under slippery conditions. Reference [6] discussed a method for improving the steering wheel returnability in an EPS system, but it focused on the return control of EPS instead of assist control. Reference [7] considered both EPS assist control and return control in detail and formulated a return control algorithm based on a time-varying sliding mode control strategy, but it did not discuss determination of the current compensation coefficient.

In this paper, a novel EPS system control strategy based on motor current compensation is developed along with a fuzzy logic algorithm that evaluates tire-road friction coefficient and vehicle speed to determine the current compensation coefficient. When road friction decreases, the controller increases steering effort to improve driving safety based on a real-time estimation of the tire-road friction coefficient. Moreover, as road friction gets smaller, steering effort is augmented further to feed back changes in road friction to the driver. The definition of steering effort augmentation factor Δ is

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explained later. Vehicle speed also influences the amount of compensation. Though it is possible for an experienced driver to sense a drop in steering effort at low vehicle speeds and react accordingly, this is much more difficult at higher speeds where human reaction is easily overwhelmed, thus necessitating additional software compensation. Not only does the proposed control algorithm enhance driving safety, but it improves road feel as well.

SYSTEM MODELING

Steering System Dynamics

An EPS is a multi-body system and modeling all parts of an EPS can be complicated and unnecessary [5]. Thus a simplified EPS model is created and depicted in Figure 1. Numerous scholars have worked on the modeling of EPS systems, so the modeling process itself will not be explained in detail. The equilibrium equations of the steering system can be defined as follows.

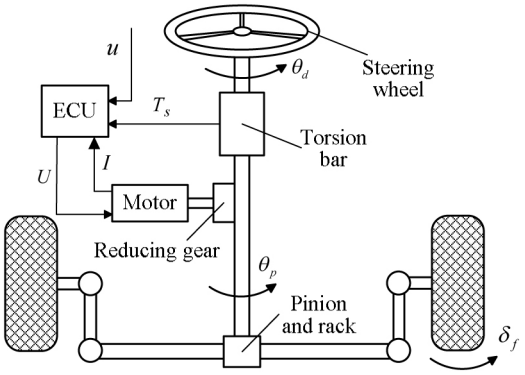


Figure 1. Steering mechanism with EPS

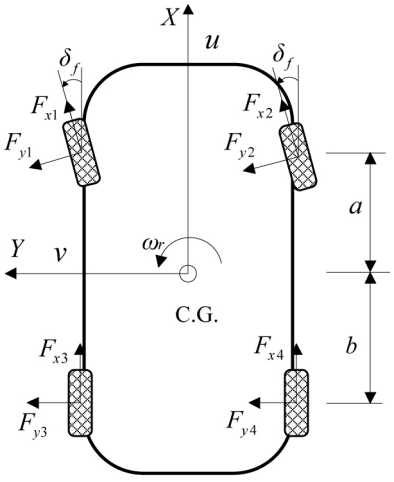


Figure 2. Vehicle model coordinates

$$J_s\ddot{\theta}_d + B_s\dot{\theta}_d + T_s = T_d \tag{1}$$

$$J_e\ddot{\theta}_p + B_e\dot{\theta}_p + T_r = T_s + T_a \tag{2}$$

Where $T_s = K_s(\theta_d - \theta_p)$, $J_e = J_p + m_r r_p^2$, $B_e = B_p + B_r r_p^2$, $\theta_p = G_e\delta_f$

The actuator utilized is a DC motor, the equilibrium equations of which can be written as follows.

$$U = RI + L\dot{I} + K_b\dot{\theta}_m \tag{3}$$

$$T_a = G_m K_z I \tag{4}$$

$$\theta_m = G_m\theta_p \tag{5}$$

The definition and value of each parameter can be found in the appendix.

Vehicle Dynamics

The vehicle model presented in this paper is a simplified model in which vehicle velocity along X axis is assumed to be constant and cross wind, roll motion and pitch motion are neglected. The vehicle model is depicted in Figure 2 with the chassis-based coordinate system fixed to the center of gravity of the vehicle.

Through force analysis, three equations can be derived as follows.

$$\begin{cases} F_x = m_v(\dot{u} - \omega_r v) \\ F_y = m_v(\dot{v} + \omega_r u) \\ M_z = I_z\dot{\omega}_r \end{cases} \tag{6}$$

Where F_x and F_y are the generalized external forces acting along the X and Y axes, respectively, M_z is the generalized moment around the Z axis, m_v is the vehicle mass, u and v are the vehicle velocity along the X and Y axes, respectively, ω_r is the yaw rate, and I_z is the moment of inertia around the Z axis perpendicular to the page.

It is through the tires that forces generated from roads act on the vehicle, so force analysis of the tires is very important. The following equation can be derived through tires force analysis.

$$\begin{cases} F_x = (F_{x1}+F_{x2})\cos\delta_f - (F_{y1}+F_{y2})\sin\delta_f + F_{x3} + F_{x4} \\ F_y = (F_{x1}+F_{x2})\sin\delta_f + (F_{y1}+F_{y2})\cos\delta_f + F_{y3} + F_{y4} \\ M_z = a(F_{x1}+F_{x2})\sin\delta_f + a(F_{y1}+F_{y2})\cos\delta_f - b(F_{y3}+F_{y4}) \end{cases} \tag{7}$$

Where a is the distance of C.G. from the front axle, b is the distance of C.G. from the rear axle, the subscripts (1, 2, 3, 4) represent the front left, front right, rear left and rear right wheels, respectively.

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The rotational speed ω_w for non-driving wheels is calculated in this section for the purpose of estimating road-tire friction coefficient below.

$$\omega_w = u / r \tag{8}$$

Where r denotes the effective radius of the wheel.

Tire Model

For the purposes of this research, a simple linear tire model is insufficient for reflecting the influence of friction coefficient and front axle load. Therefore, we adopted a Dugoff tire model [8] to illustrate the nonlinear characteristic of the tire. The Dugoff tire model allows us to calculate longitudinal and lateral forces of each tire. Its mathematical equations are as follows.

$$\begin{cases} F_{xi} = C_{si} s_i f(\lambda_i)/(1+s_i) \\ F_{yi} = C_{\alpha i} \tan\alpha_i f(\lambda_i)/(1+s_i) \end{cases} \tag{9}$$

Where the subscript i = 1, 2, 3, 4, C_{si} and $C_{\alpha i}$ represent the longitudinal and lateral tire stiffness, respectively. The variable $f(\lambda_i)$ can be calculated by the following equations.

$$\lambda_i = \frac{\mu F_{zi}(1+s_i)}{2\sqrt{(C_{si}s_i)^2 + (C_{\alpha i}\tan\alpha_i)^2}} \tag{10}$$

$$f(\lambda_i) = \begin{cases} (2-\lambda_i)\lambda_i & \lambda_i < 1 \\ 1 & \lambda_i \geq 1 \end{cases} \tag{11}$$

Where, μ is the tire-road friction coefficient, F_{zi} denotes the normal load of each road wheel calculated using equation (12), S_i is the longitudinal slip rate and α_i is the slip angle calculated using equation (13).

$$\begin{cases} F_{z1} = \frac{mgb}{2(a+b)} - \frac{m_v(\dot{u}-\omega_r v)h}{2(a+b)} - \frac{m_v(\dot{v}+\omega_r u)hb}{T_f(a+b)} \\ F_{z2} = \frac{mgb}{2(a+b)} - \frac{m_v(\dot{u}-\omega_r v)h}{2(a+b)} + \frac{m_v(\dot{v}+\omega_r u)hb}{T_f(a+b)} \\ F_{z3} = \frac{mga}{2(a+b)} + \frac{m_v(\dot{u}-\omega_r v)h}{2(a+b)} - \frac{m_v(\dot{v}+\omega_r u)ha}{T_r(a+b)} \\ F_{z4} = \frac{mga}{2(a+b)} + \frac{m_v(\dot{u}-\omega_r v)h}{2(a+b)} + \frac{m_v(\dot{v}+\omega_r u)ha}{T_r(a+b)} \end{cases} \tag{12}$$

$$\begin{cases} \alpha_1 = \delta_f - \arctan(\frac{a\omega_r + v}{u - 0.5T_f\omega_r}) \\ \alpha_2 = \delta_f - \arctan(\frac{a\omega_r + v}{u + 0.5T_f\omega_r}) \\ \alpha_3 = \arctan(\frac{b\omega_r - v}{u - 0.5T_r\omega_r}) \\ \alpha_4 = \arctan(\frac{b\omega_r - v}{u + 0.5T_r\omega_r}) \end{cases} \tag{13}$$

Where T_f and T_r are respectively the front track and rear track, h is the height of C.G. from the ground, g is the acceleration of gravity.

Calculation of Steering Resistance Torque

During a steering maneuver, the resistance torque comprises three components, namely self-alignment torque M_{f1}, gravity alignment torque M_{f2} and friction torque M_{f3}. Tire deformation at the front part of the contact patch is greater than at the rear, giving rise to uneven distribution of cornering force which generates self-alignment torque. M_{f1} can be defined as:

$$M_{f1} = (F_{y1} + F_{y2})(t_m + t_p) \tag{14}$$

Where t_m and t_p are the mechanical trail and pneumatic trail, respectively.

The gravity alignment torque, M_{f2}, results from the geometry of the steering system. It can be expressed as follows [9].

$$M_{f2} = (F_{y1} + F_{y2})(c + r\tan\varphi)\cos\varphi\sin\varphi\sin\delta_{fm} \tag{15}$$

Where c is the kingpin offset distance, φ is the kingpin inclination angle, and δ_{fm} is the average steering angle of the two front wheels. In this paper $\delta_{fm} = \delta_f$.

The friction torque, M_{f3}, results from friction within the tire-road interface and the steering system and can be measured by experiment [7]. The total resistance torque at the steering column can be calculated as follows:

$$T_r = (M_{f1} + M_{f2} + M_{f3})/G_e \tag{16}$$

Note that the equations presented here describe decentering steering torques only, as opposed to recentering torques, which require a different set of equations.

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CONTROL STRATEGY

Estimation of Tire-Road Friction Coefficient

The estimation of tire-road friction coefficient is a prerequisite to formulating the control strategy. Researches in this field focus mainly on vehicle acceleration and braking cases [10, 11], which are not applicable in this paper. To accommodate steering cases, we adopt a model-based estimation algorithm proposed by Taehyun Shim and Donald Margolis [12], so that tire-road friction coefficient can be estimated from easily measured signals (e.g. yaw rate and wheel rotation speed). The schematic diagram of the friction coefficient estimation algorithm is shown in Figure 3.

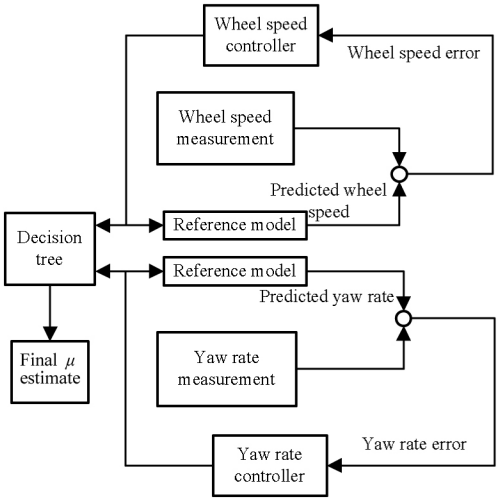
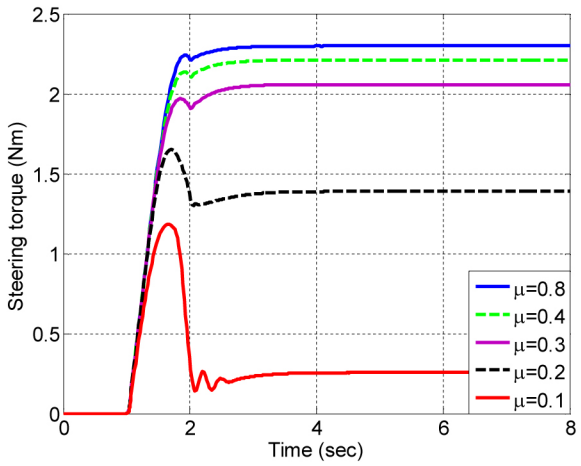


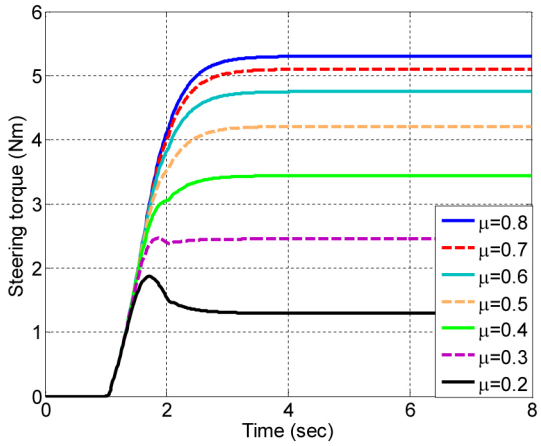
Figure 3. Schematic diagram of μ estimation algorithm

Control Algorithm

Since the output torque of a DC motor is proportional to its input current, we can control the current to produce the ideal assist motor torque. In a conventional EPS system, the ideal assist motor current is determined by a steering characteristic curve. In other words, when the driver applies a torque to the steering wheel at a certain vehicle speed, the ideal assist motor current can be determined uniquely. This type of control strategy, however, does not consider the influence of other factors (e.g. tire-road friction coefficient) that are also very important for safety and road feel. Figure 4 shows the influence of tire-road friction coefficient on the steering wheel torque at two vehicle speeds under a ramp-step steering input of 90 degrees in the low speed case and 45 degrees in the high speed case. Note that the friction coefficient exerts greater influence on steering effort with increasing vehicle speed. Hence, when driving on slippery roads, steering may feel unusual or awkward with a conventional EPS system.



(a). Vehicle speed u=30km/h



(b). Vehicle speed u=60km/h

Figure 4. The influence of friction coefficient on steering effort

To improve EPS design, this paper proposes a novel control strategy that adds a compensation current to the basic motor input current. The compensation current is closely associated with tire-road friction coefficient, and its gain coefficient is calculated by a fuzzy logic controller. The key requirements of the compensation current controller are as follows:

1. The lower the friction coefficient is, the less assist torque the motor should provide, so that required steering effort is increased. The reduction in assist torque can be achieved by adding a negative compensation current.
2. At a certain vehicle speed, the required steering effort on slippery roads should be made higher than that on normal roads, so as to alert the driver to a drop in road friction. Under a ramp step steering input, assuming the steady state value of steering torque on normal roads is T_s and the steady state value of augmented steering torque on slippery roads is $T^*_{s\mu}$, we can define the steering torque augmentation factor Δ, which is a vital evaluation criterion representing the amount of feedback provided to the driver when road friction changes. Δ can be written as follows.

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$$\Delta = \frac{T_{s\mu}^{*} - T_s}{T_s} \times 100\% \tag{17}$$

To provide the necessary feedback, Δ should increase with decreasing road friction. According to reference [13], Δ can be determined from the driver's requirements and is typically within the range of 0 < Δ < 1, although we shall limit Δ to 0.25 to avoid causing excessive steering effort.

3. As can be seen in Figure 4, at low vehicle speeds the drop in steering effort when transitioning from normal to slippery roads is small, and ease of steering, rather than stability, should be the priority, thus the reduction of motor assist torque in this case should be correspondingly small. On the other hand, steering effort decreases considerably at higher speeds, thus greater reduction in motor assist torque is required to increase steering effort. Interestingly, when vehicle speed is above a critical value, the ideal motor assist torque can become negative, that is, the motor needs to generate a resist torque to bring steering effort to required levels.
4. With the increase of steering wheel angle, the change of the steering torque could be larger, so the compensation current should also be increased.
5. The compensation control should not be active at all times. This is because more time is usually spent driving at low and medium speeds under high *μ* conditions, in which case the capabilities of a conventional EPS are sufficient. The proposed current compensation strategy is more effective under severe conditions.

In light of all the requirements stated above, the proposed control algorithm can be defined as follows.

$$I_i = I_b - I_c \tag{18}$$

$$I_c = K\theta_d \tag{19}$$

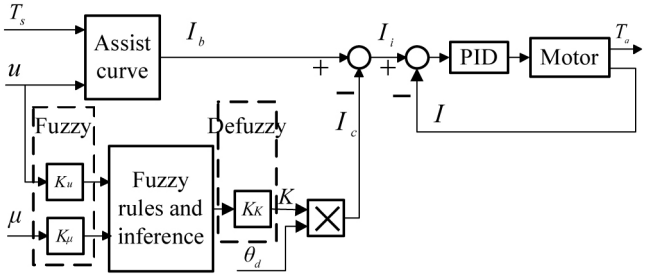


Figure 5. Overall control structure

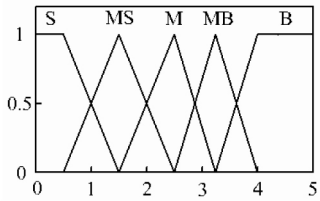


Figure 6. Membership functions for friction coefficient

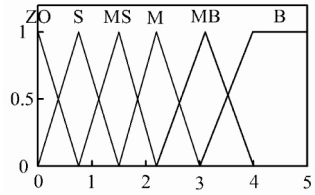


Figure 7. Membership functions for vehicle speed

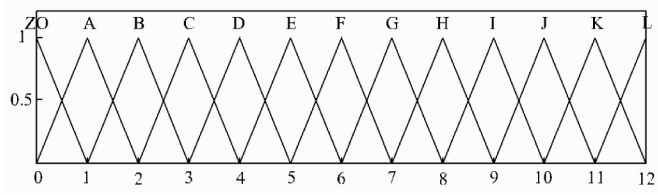


Figure 8. Membership functions for compensation coefficient

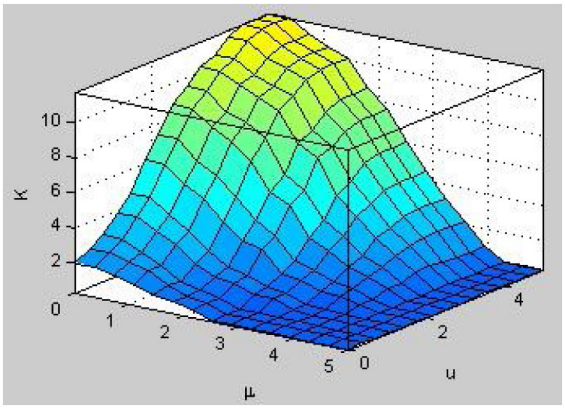


Figure 9. Three-dimensional fuzzy relationship

Table 1. Fuzzy rules

K		u					
		ZO	S	MS	M	MB	B
μ	S	B	C	F	H	J	L
	MS	A	A	B	E	H	J
	M	ZO	ZO	A	B	D	F
	MB	ZO	ZO	ZO	A	B	C
	B	ZO	ZO	ZO	ZO	ZO	ZO

Where I_i, I_b and I_c denote the ideal motor input current, base input current, and compensation current, respectively, and K is the compensation coefficient. Note that K is not a constant as it increases with decreasing friction coefficient and increasing vehicle speed. The structure of the proposed cascade control

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strategy can be seen in Figure 5. The base input current I_b can be calculated using torsion bar and vehicle speed signals in the assist curve block, the formulation of which can be found in relevant literature. Moreover, a fuzzy logic control strategy, to be explained below, is utilized to determine the value of K.

The friction coefficient μ and vehicle speed u are fuzzy logic input variables while the coefficient of current compensation K is the output variable. The range of μ, u and K are defined as (0, 1], [0, 100km/h] and [0, 12] respectively. Additionally, K_μ and K_u are input scaling factors, which have values of 5 and 0.05 respectively. K_K is the output scaling factor which equals to 1. Following the definition of input and output variables and their ranges, membership functions (Figures 6, 7, 8) are built for each. For control accuracy, the membership of K is divided into 13 levels. It incorporates a zero level, which represents no current compensation. We can then design the rules and inferences based on the key controller requirements as mentioned above. The fuzzy inference is of Mamdani-type. Fuzzy rules are explicitly written in table 1, and the three-dimensional fuzzy relationship of μ, u and K is shown in Figure 9.

EXPERIMENTAL AND SIMULATION RESULTS

Experimental Validation of Model

The overall model of an EPS-equipped vehicle, shown in Figure 10, is built and analyzed in MATLAB/Simulink. Vehicle parameters are summarized in table 2.

Physical tests were conducted to validate the model. The experiment setup is shown in Figure 11. As can be seen, a steering sensor (SG-310) was installed on the steering wheel to gather signals for steering wheel angle and torque. These signals were transmitted to a mobile data recorder (MDR) for data processing. Additionally a computer was set up to monitor and record experiment data.

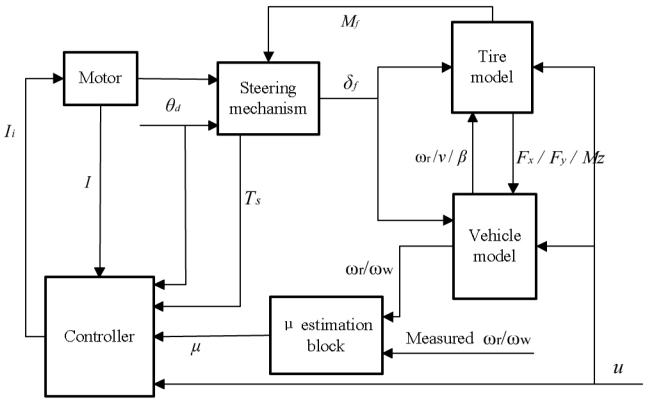
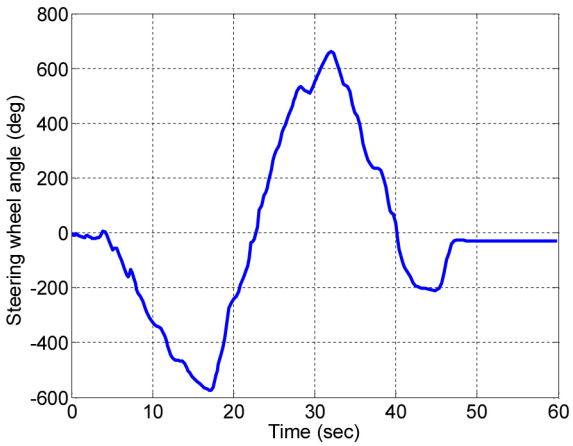


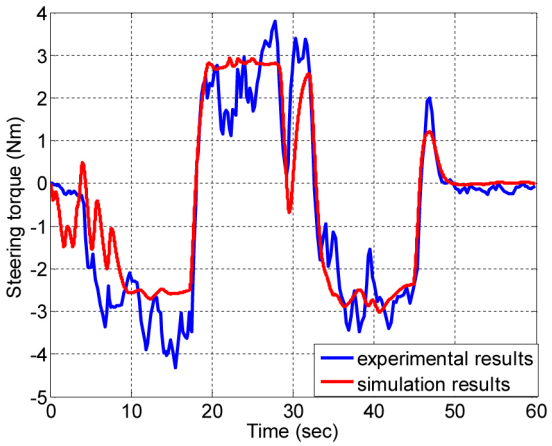
Figure 10. The Simulink model of a vehicle with EPS



Figure 11. Experimental setup



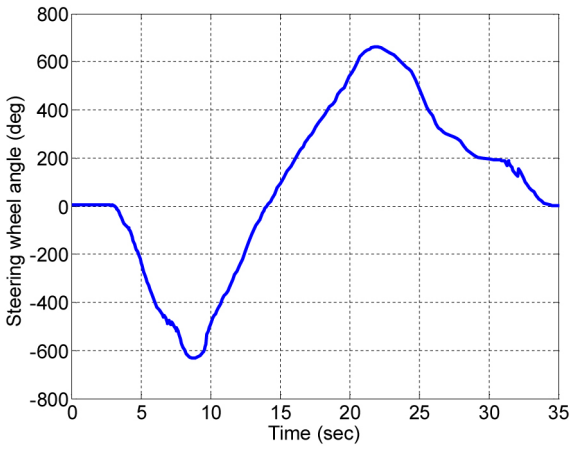
(a). Steering wheel angle input



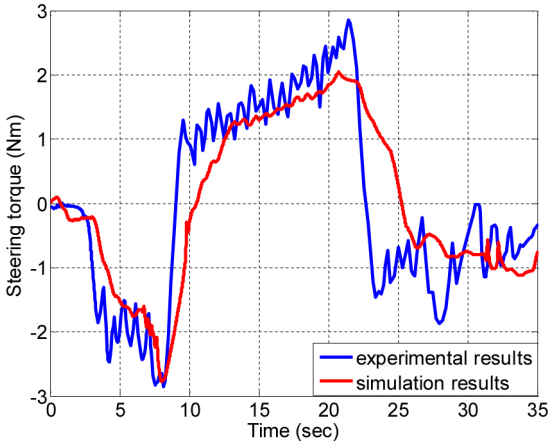
(b). Steering torque comparison

Figure 12. Tests on dry asphalt

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(a). Steering wheel angle input



(b). Steering torque comparison

Figure 13. Tests on snow-covered road

Table 2. Vehicle parameters

Parameters	Values
Vehicle mass	1150 kg
Moment of inertia around the Z axis	2007 kg * m2
Distance of C.G. from the front axle	1.248 m
Distance of C.G. from the rear axle	1.357 m
Height of C.G. from the ground	0.58m
Wheel base	2.605 m
Front track	1.47m
Rear track	1.46m
Tire size	195/60R15

For driving safety, the experiments were carried out when steering on the spot. Results from a dry asphalt road test are shown in Figure 12, with steering wheel angle displayed in Figure 12 (a) and the corresponding steering effort from both experiment and simulation shown in Figure 12 (b). Test results for a snow-covered road are shown in Figure 13, with steering wheel angle depicted in Figure 13 (a) and steering effort shown in Figure 13 (b). Simulation results were found to be in good agreement with experimental results, demonstrating that the vehicle model built in MATLAB/Simulink is suitable for controller design. Moreover, the larger mean square value of driver steering torque on dry asphalt roads compared to that on snow-covered roads demonstrates the central premise of this paper and thus the necessity for improved road feel when driving on low μ roads.

Simulation Results

Simulations were conducted to determine steering torques with and without the proposed compensation strategy in low speed (30km/h) and high speed (60km/h) scenarios using a ramp-step steering input of amplitude 90 degrees at low speed and 45 degrees at high speed. Results are shown in Figures 14 and 15 where the red, green and blue curves represent the steering effort on normal roads, slippery roads with no compensation and slippery roads with compensation, respectively. To avoid clutter, only test data for μ = 0.3 roads is shown graphically.

In a similar manner, we can examine steering effort for a range of friction coefficients under typical driving conditions [14]. The effect of compensation at low vehicle speed can be seen in table 3 below. Here, the steady state value of steering torque on slippery roads with no compensation is denoted $T'_{s\mu}$, while $T^*_{s\mu}$ and Δ retain their definitions from sections above. For reasons previously explained, no compensation was used in low vehicle speed and high μ scenarios. The effect of compensation in high speed scenarios is shown in table 4. A high speed, icy road scenario is not considered at this point because driving the car on icy roads at high speed was deemed impractical.

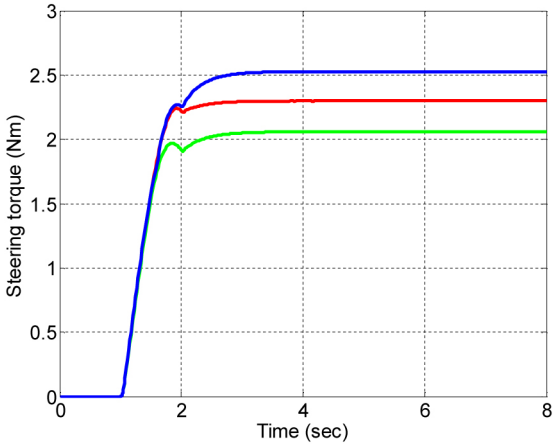


Figure 14. Simulation at low vehicle speed

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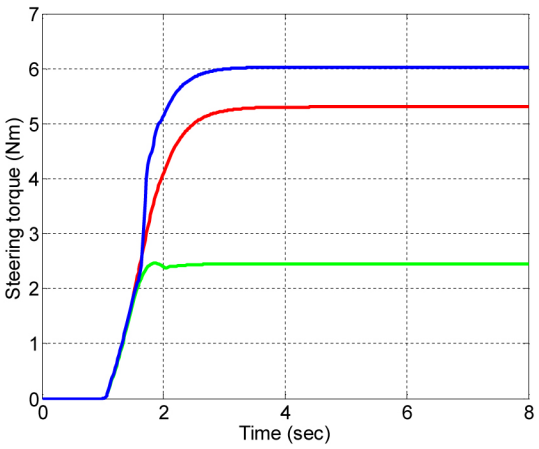


Figure 15. Simulation at high vehicle speed

Table 3. Simulation results at low vehicle speed

Conditions	μ	$T_{s\mu}^{'}$ (Nm)	$T_{s\mu}^{*}$ (Nm)	Δ
Dry asphalt	0.8	2.30	2.30	0
Wet asphalt	0.5	2.21	2.44	6.09%
Snowy roads	0.2	1.39	2.61	13.48%
Icy roads	0.1	0.26	2.74	19.13%

Table 4. Simulation results at high vehicle speed

Conditions	μ	$T_{s\mu}^{'}$ (Nm)	$T_{s\mu}^{*}$ (Nm)	Δ
Dry asphalt	0.8	5.30	5.30	0
Dry soil	0.68	5.05	5.49	3.58%
Cobblestone	0.6	4.75	5.62	6.04%
Wet asphalt	0.5	4.21	5.97	12.64%
Snowy roads	0.2	1.30	6.23	17.55%

In low speed scenarios, a decrease in road friction has little influence on vehicle handling, whereas in high speed scenarios, the impact is considerable. Thus, the amount of steering effort added increases with vehicle speed. In addition, the steering effort augmentation factor Δ also increases with decreasing road friction to feed back changes in road friction to the driver. Note that the Δ in the proposed EPS system is limited to 25% so as not to compromise ease of steering.

CONCLUSION

Physical testing has validated the numerical model built in MATLAB/Simulink and confirmed the central premise that steering effort on high μ roads is larger than that on low μ roads. Simulation results showed that the proposed current compensation control strategy is able to increase steering effort to appropriate values when driving on slippery roads such that steering effort increases with decreasing μ so as to alert the driver of decreasing road friction. With the new EPS system, the driver is unlikely to steer excessively on slippery roads, resulting in improved safety and road feel.

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DEFINITIONS/ABBREVIATIONS

Parameters	Description	Values
J_s	Moment of inertial of steering wheel and input column	0.017 $kg \cdot m^2$
J_e	Equivalent moment of inertial reflected to output column	
J_p	Moment of inertial of output column and gear box	0.022 $kg \cdot m^2$

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m_r	Mass of rack and front road wheels	35 *kg*
B_s	Damping coefficient of steering wheel and input column	0.13 $N \cdot m \cdot s/rad$
B_e	Equivalent damping coefficient reflected to output column	
B_p	Damping coefficient of output column and gear box	0.21 $N \cdot m \cdot s/rad$
B_r	Damping coefficient of rack	0.12 $N \cdot m \cdot s/rad$
θ_d	Steering wheel angle	
θ_p	Output column angle	
δ_f	Front road wheel angle	
T_d	Steering wheel torque	
T_s	Torque measured by torsion bar	
T_a	Motor assistant torque	
T_r	Resistant torque	
K_s	Stiffness of torsion bar	139 $N \cdot m/rad$
r_p	Radius of pinion	0.0074 *m*
G_e	Transmission ratio of pinion and rack	18
G_m	Transmission ratio of gear box	16.5
U	Motor control voltage	
I	Motor actual current	
R	Motor armature resistance	0.06 Ω
L	Motor armature inductance	0.0001 *H*
K_b	Back EMF coefficient	0.065 $V \cdot s/rad$
K_z	Torque constant of the motor	0.065 $N \cdot m/A$

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